

10	A	The gravitational potential at the surface of a planet is directly proportional to the planet's mass, and inversely proportional to its radius (or diameter). Hence, the potential at Mars' surface is approximately $(0.1 / 0.5) \times (-63 \text{ MJ kg}^{-1}) = -13 \text{ MJ kg}^{-1}$.
11	B	This is an extension from Q. 10, considering the rate of thermal energy removed